

```

clc;
clear;
close all;

%% MATLAB-Based Modelling and Performance Analysis of AVR
% Student Mini-Project
% Model: Amplifier + Exciter + Generator + Sensor

%% 1. Define AVR Parameters
Ka = 10;          % Amplifier gain
Ta = 0.1;         % Amplifier time constant (s)

Ke = 1;           % Exciter gain
Te = 0.4;         % Exciter time constant (s)

Kg = 1;           % Generator gain
Tg = 1.0;         % Generator time constant (s)

Ks = 1;           % Sensor gain
Ts = 0.01;        % Sensor time constant (s)

%% 2. Define the Laplace Variable
s = tf('s');

%% 3. Model Each AVR Component
G_amplifier = Ka / (Ta*s + 1);
G_exciter   = Ke / (Te*s + 1);
G_generator = Kg / (Tg*s + 1);
H_sensor    = Ks / (Ts*s + 1);

%% 4. Form Open-Loop and Closed-Loop Models
G_open = G_amplifier * G_exciter * G_generator;
G_AVR = feedback(G_open, H_sensor);

disp('Open-loop transfer function:');

```

Open-loop transfer function:

G_open

```

G_open =
          10
-----
0.04 s^3 + 0.54 s^2 + 1.5 s + 1

```

Continuous-time transfer function.
Model Properties

```

disp('Closed-loop AVR transfer function:');

```

Closed-loop AVR transfer function:

G_AVR

```
G_AVR =  
          0.1 s + 10  
-----  
0.0004 s^4 + 0.0454 s^3 + 0.555 s^2 + 1.51 s + 11
```

Continuous-time transfer function.

Model Properties

```
%% 5. Check Stability  
poles_AVR = pole(G_AVR);  
  
disp('Closed-loop poles:');
```

Closed-loop poles:

```
disp(poles_AVR);  
  
-99.9712 + 0.0000i  
-12.4892 + 0.0000i  
-0.5198 + 4.6642i  
-0.5198 - 4.6642i
```

```
if isstable(G_AVR)  
    disp('Result: AVR model is stable.');
```

else

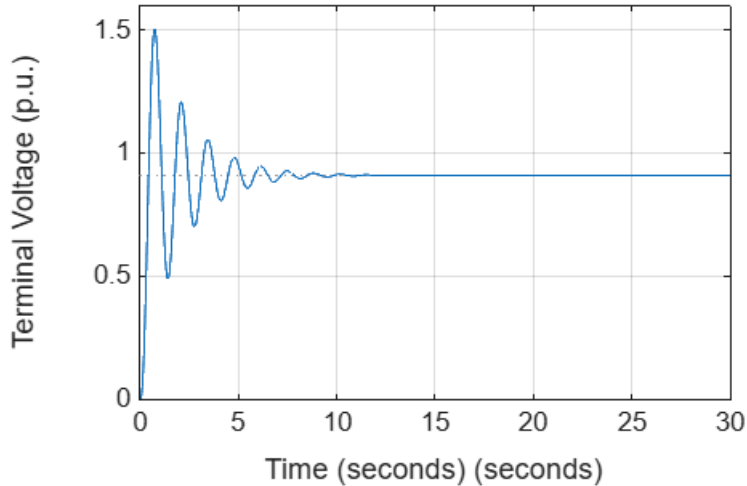
```
    disp('Result: AVR model is unstable.');
```

end

Result: AVR model is stable.

```
%% 6. Plot Step Response  
figure('Name', 'AVR Step Response');  
step(G_AVR, 30);  
grid on;  
title('Closed-Loop AVR Step Response');  
xlabel('Time (seconds)');  
ylabel('Terminal Voltage (p.u.)');
```

Closed-Loop AVR Step Response



%% 7. Calculate Performance Parameters

```
final_voltage = dcgain(G_AVR);
steady_state_error = abs(1 - final_voltage) * 100;
```

```
fprintf('\n--- AVR Performance ---\n');
```

```
--- AVR Performance ---
```

```
fprintf('Final voltage: %.4f p.u.\n', final_voltage);
```

```
Final voltage: 0.9091 p.u.
```

```
fprintf('Steady-state error: %.4f %%\n', steady_state_error);
```

```
Steady-state error: 9.0909 %
```

```
if isstable(G_AVR)
    info = stepinfo(G_AVR);

    fprintf('Rise time: %.4f seconds\n', info.RiseTime);
    fprintf('Settling time: %.4f seconds\n', info.SettlingTime);
    fprintf('Overshoot: %.4f %%\n', info.Overshoot);
else
    disp('Rise time and settling time are not reported for an unstable
model.');
```

```
end

Rise time: 0.2607 seconds
Settling time: 6.9865 seconds
Overshoot: 65.7226 %
```

%% 8. Compare Different Amplifier Gains

```

Ka_values = [2 5 10];

RiseTime = NaN(length(Ka_values),1);
SettlingTime = NaN(length(Ka_values),1);
Overshoot = NaN(length(Ka_values),1);
FinalVoltage = NaN(length(Ka_values),1);
Stable = false(length(Ka_values),1);

figure('Name','Comparison of Amplifier Gains');
hold on;

for i = 1:length(Ka_values)

    Ka_test = Ka_values(i);

    G_amp_test = Ka_test / (Ta*s + 1);
    G_open_test = G_amp_test * G_exciter * G_generator;
    G_test = feedback(G_open_test, H_sensor);

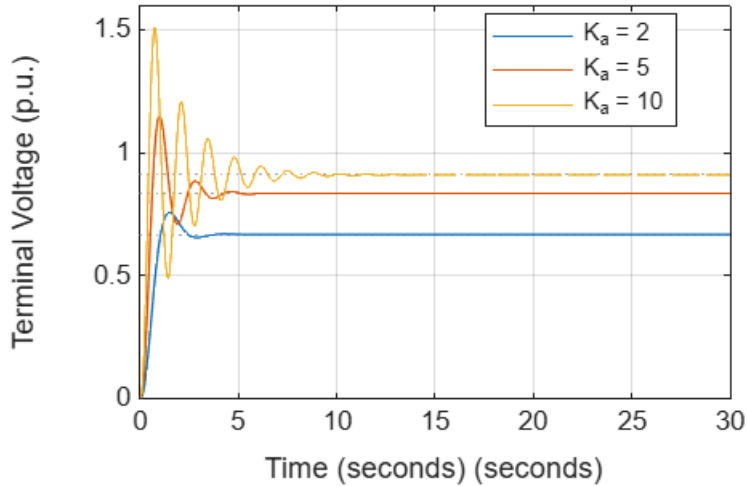
    % Plot only stable responses
    if isstable(G_test)
        step(G_test, 30);
        Stable(i) = true;

        info_test = stepinfo(G_test);
        RiseTime(i) = info_test.RiseTime;
        SettlingTime(i) = info_test.SettlingTime;
        Overshoot(i) = info_test.Overshoot;
        FinalVoltage(i) = dcgain(G_test);
    else
        FinalVoltage(i) = dcgain(G_test);
    end
end

grid on;
title('AVR Response for Different Amplifier Gains');
xlabel('Time (seconds)');
ylabel('Terminal Voltage (p.u.)');
legend('K_a = 2', 'K_a = 5', 'K_a = 10', ...
        'Location','best');
hold off;

```

AVR Response for Different Amplifier Gains



%% 9. Display Comparison Table

```
Results = table(Ka_values', Stable, RiseTime, SettlingTime, ...
    Overshoot, FinalVoltage, ...
    'VariableNames', {'AmplifierGain', 'Stable', ...
    'RiseTime_s', 'SettlingTime_s', ...
    'Overshoot_percent', 'FinalVoltage_pu'});

disp(' ');
```

```
disp('--- AVR Performance Comparison Table ---');
```

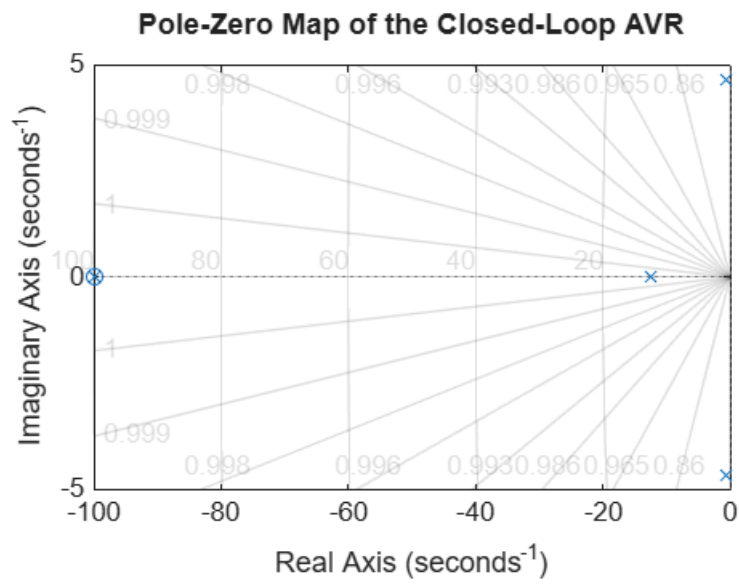
```
--- AVR Performance Comparison Table ---
```

```
disp(Results);
```

AmplifierGain	Stable	RiseTime_s	SettlingTime_s	Overshoot_percent	FinalVoltage_pu
2	true	0.66773	2.2828	13.588	0.66667
5	true	0.3858	3.895	38.01	0.83333
10	true	0.26074	6.9865	65.723	0.90909

%% 10. Plot Closed-Loop Poles for the Main Model

```
figure('Name', 'AVR Pole Plot');
pzmap(G_AVR);
grid on;
title('Pole-Zero Map of the Closed-Loop AVR');
```



```
disp('Project simulation completed.');
```

Project simulation completed.